Shivam Kamboj

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SUMMARY

Quantum information researcher (Physics Ph.D., UC Merced, anticipated Fall 2026) working across the quantum simulation of many-body systems, quantum error correction, and quantum-hardware characterization. My thesis models coupled quantum Rabi arrays — ultrastrong light–matter systems — using large-scale exact diagonalization (QuTiP, HPC) to resolve quantum phase transitions, dynamics, and symmetry-protected Schrödinger-cat ground states. Alongside the theory I built and released an open-source library benchmarking surface-code error correction under correlated, non-Markovian noise, and, at Lawrence Livermore National Lab, developed pulse-level control and process-tomography software for superconducting qudits. First-author theory published in *Physical Review B*, hands-on collaboration across theory and experiment, and a consistent practice of reproducible, well-engineered scientific software. Interested in research-scientist and applied-scientist roles spanning QEC, quantum simulation, and quantum applications.

TECHNICAL SKILLS

Quantum Simulation & Many-Body Physics: Hamiltonian simulation and exact diagonalization; quantum phase transitions and crossover phenomena; symmetry- and parity-protected ground states; Schrödinger-cat / bosonic states and Wigner-function tomography; ultrastrongly coupled cavity-QED, Jaynes–Cummings / quantum Rabi models; open quantum systems / master-equation dynamics; Bogoliubov and normal-mode analysis

Quantum Error Correction: Surface-code error correction and stabilizer simulation (Stim, PyMatching), MWPM decoding, Monte-Carlo threshold estimation with bootstrap confidence intervals, correlated / non-Markovian ($1/f$) noise modeling, Ornstein–Uhlenbeck trajectory methods

Scientific Computing & HPC: Sparse linear algebra (SciPy / ARPACK, QuTiP), large-operator exponentiation, performance and memory profiling / optimization, parallel computing (multiprocessing, Joblib), SLURM scheduling, checkpointed long-running jobs at $\sim 10^4$ Hilbert-space dimension on Intel Xeon clusters

Quantum Hardware & Control: Superconducting qubits/qudits (circuit-QED), photonic / cavity-QED, and magnon–photon (spin–cavity) platforms; pulse-level quantum control, gate calibration and optimization, quantum process tomography, qubit characterization (T1, T2, Ramsey)

Programming & Software Engineering: Python (NumPy, SciPy, Pandas, Matplotlib, Flask), Qiskit, Stim, PyMatching, QuTiP, TensorFlow / Keras; reproducible research engineering (`pytest`, `ruff`, `mypy -strict`, continuous integration); Git/GitHub, Linux, Bash, Google Cloud, LaTeX

Communication & Collaboration: First-author publication and technical writing; conference talks and posters (APS, national lab); invited seminars and guest lectures; theory–experiment collaboration; two-time Outstanding Teaching Award recipient

RESEARCH & PROFESSIONAL EXPERIENCE

University of California, Merced — *Merced, CA*
Graduate Researcher — Tian Group (Theoretical Quantum Optics / Quantum Computing) — May 2023 – Present

- Lead investigator on my doctoral research, *Quantum Phase Transitions and Dynamical Behavior in Coupled Quantum Rabi Arrays*: designed and maintain a QuTiP / Python codebase (sparse exact diagonalization, open quantum systems) mapping ultrastrong light–matter Hamiltonians to their spectra, dynamics, and quantum-phase-transition structure. First-author manuscript in preparation.
- Identified and characterized a crossover window hosting symmetry- (parity-) protected Schrödinger-cat ground states, with Wigner-function tomography across the trusted parameter regime and boundary validation against multiple analytical and numerical methods.
- Engineered the simulations as large-scale HPC pipelines on the UC Merced Pinnacles cluster (SLURM, checkpointing, memory / worker-count tuning for large-operator exponentiation) at Hilbert-space dimensions $\sim 10^4$; documented a trusted-regime convergence analysis separating physical divergence from numerical artifact.

Lawrence Livermore National Laboratory (LLNL) — *Livermore, CA*
Physics Graduate Intern — Quantum Coherent Device Physics Group — May 2024 – Aug 2024

- Developed pulse-level quantum control software in Python for superconducting qudit hardware; designed, benchmarked, and optimized gate sequences with fidelity improvements validated end-to-end via quantum process tomography.

- Engineered a generalized, linear-inversion quantum process tomography pipeline (pseudo-inverse reconstruction) to recover process maps from noisy experimental data, delivering rapid feedback for automated device characterization.
- Automated device-coherence characterization by fitting Ramsey fringes and T1/T2 decay, closing the loop between simulated device models and measured hardware behavior.

Collaborator — Sharping Group (Experimental Quantum Optics) 2024 – Present

- Model a YIG–cavity dimer realized in a magnon–photon platform, demonstrating that the same modeling methods transfer across distinct physical implementations and adding a third qubit-architecture family to my hands-on experience.

San Jose State University & UC Merced *San Jose / Merced, CA*
Graduate Researcher — Hurst Group Jun 2021 – Jan 2024

- First-author theoretical study of oscillatory edge modes in 2D spin-torque oscillator arrays: built a Python simulation framework, derived analytic dispersion relations, and demonstrated topological robustness against disorder (see Publications).

SELECTED PROJECTS

`nonmarkov_qec` *Python, NumPy/SciPy, Stim, PyMatching*
Open-source library for benchmarking QEC under non-Markovian noise — v1 released 2025 – Present

- Designed, built, and released a full surface-code QEC benchmarking pipeline: Ornstein–Uhlenbeck / sum-of-OU generation of correlated $1/f$ noise, Stim stabilizer simulation, MWPM/PyMatching decoding, and two-layer Monte-Carlo threshold estimation with bootstrap confidence intervals; validated on rotated surface codes at distances $d = 3$ and $d = 5$.
- Established threshold invariance across noise correlation times $\tau_c = 0.2$–200 cycles ($p_{\text{th}} \approx 3.3 \times 10^{-3}$), independently reproducing a published correlated-noise result. Engineered as reproducible software: 74-test suite, `ruff` + `mypy -strict` clean, continuous integration.

Physics Concept Analyzer Tool (PCAT) *Python, Flask, Gemini API, GCP*
Full-stack, deployed application — in active production use at UC Merced 2025

- Designed, built, and deployed an end-user application end-to-end on Flask + Google App Engine, taking it from specification to production and real adoption; integrated a structured multimodal LLM pipeline (Google Gemini).

PUBLICATIONS

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Physical Review B* **109**, 094436 (2024). DOI: 10.1103/PhysRevB.109.094436
2. S. Kamboj and L. Tian, "Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer," manuscript in preparation, 2026.

EDUCATION

University of California, Merced *Merced, CA*
Ph.D. in Physics — Quantum Computing (Advisor: Prof. Lin Tian) Anticipated Fall 2026

Relevant coursework: Computational Physics, Advanced Quantum Computing

San Jose State University (SJSU) *San Jose, CA*
M.S. in Physics 2022

Relevant coursework: Quantum physics, Optics, Methods in mathematical physics, Machine learning in astronomy

AWARDS, PRESENTATIONS & OUTREACH

- **Outstanding Teaching Assistant Award**, UC Merced (2026) and San Jose State University (2022) — two-time recipient across both graduate institutions.
- Physics Graduate Group Summer Fellowship ($7,000/summer), UC Merced — 2025 & 2026; Travel Fellowship ($750), 2026.
- Talks & posters: APS Global Physics Summit 2026 (poster); LLNL PLS Division Student Poster Symposium 2024; CSU Stanislaus Friday Seminar 2024 ("Simulating Quantum Systems"); UC Merced PHYS 281 guest lecture on Computational Physics 2024; APS March Meeting 2022 (poster).